

Report on

Determination of the Horizontal Component of Earth's Magnetic Field Using Physics-Informed Neural Networks and a Tangent Galvanometer Experiment.

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Abstract

This study presents a novel approach to estimate the horizontal component of Earth's magnetic field (BH) at NISER Bhubaneswar using Physics-Informed Neural Networks (PINNs) integrated with experimental data from a Tangent Galvanometer (TG) setup. Traditional methods rely on direct application of tangent law; however, PINNs offer a data-driven yet physics-constrained alternative. By incorporating the governing physical laws into the loss function of a neural network, we demonstrate improved consistency, accuracy, and interpretability in the estimation of BH. The results show that PINNs can effectively bridge experimental measurements and theoretical models, offering new possibilities for precision experiments in educational and research laboratories.

1. Introduction

The Earth's magnetic field has horizontal (B_h) and vertical (B_v) components, with B_h playing a crucial role in geomagnetic studies and navigation. The Tangent Galvanometer (TG) is a classic apparatus used to measure B_h by exploiting the tangent law.

$$B_h = \frac{\mu_0 n I}{2a} \tan \theta$$

Where:

- μ_0 = *permeability of free space*
- N =number of turns in the coil
- I =current through the coil
- R =radius of the coil
- θ = *deflection angle of the compass needle*

However, real-world experiments suffer from systematic errors(e.g., coil misalignment, non-uniformity in winding, and external magnetic interference), limiting accuracy.

Recent advances in Physics-Informed Neural Networks (PINNs) offer a promising solution by integrating physical laws directly into machine learning models. This work leverages PINNs to augment traditional TG experiments, combining empirical data with the underlying physics (Biot-Savart law and tangent law) to improve the accuracy and robustness of B_h estimation.

2. Experimental Setup and Data Collection

2.1 Apparatus

- **Tangent Galvanometer** : A circular coil of radius $R=0.068\text{m}$ with $N=500$ or 50 turns.
- **Compass** : Placed at the coil's center to measure deflection angle θ .
- **Power supply and ammeter** : Variable DC current (I).
- **Non- magnetic stand** : To minimize external interference.

2.2 Procedure

1. Align the TG coil in the magnetic meridian plane.
2. For each current I , record the compass deflection θ (averaged over 5 trials).
3. Repeat for counter-clockwise currents to cancel Earth's field asymmetry.

2.3 Data

Horizontal component of Earth's magnetic field for 50 turns of coil and 0.068m of radius.

Sl no	Current(mA)	Deflection of compass θ (degree)	Deflection of compass by reversing current	Average θ

			direction θ (degree)	
1	30	19	20	19.5
2	40	24	25	24.5
3	54	31	32	31.5
4	62	35	36	35.5
5	76	40	41	40.5
6	85	43	45	44
7	113	50	53	51.5
8	137	55	58	56.5
9	167	60	64	62
10	226	66	70	68

For 500 turns of coil and 0.068m of radius , Horizontal component of Earth's magnetic field.

Sl no	Current(mA)	Deflection of compass θ(degree)	Deflection of compass by reversing current direction θ (degree)	Average θ
1	3	18	18	18
2	4.2	24	24	24
3	5.6	30	30	30
4	6.9	36	36	36
5	8.8	42	43	42.5

6	10.4	47	49	48
7	13.1	53	55	54
8	16	58	60	59
9	19.3	62	65	63.5
10	25.4	67	71	69

3. Theoretical Framework

The tangent law derives from the superposition of Earth's field (B_h) and the coil's field ($B_c = \frac{\mu_0 n I}{2a}$):

$$B_c = B_h \tan \theta$$

Systematic errors arise from:

Coil imperfections (non-circularity, finite thickness).

Misalignment of the coil plane with the magnetic meridian.

External noise (e.g., electronic equipment).

4. Physics-Informed Neural Networks (PINNs)

4.1 Model Architecture

Input: Current I .

Output: Predicted deflection θ .

Hidden layers: 3 dense layers (64 neurons each, ReLU activation).

Loss function:

Data loss: MSE between predicted and observed θ .

Physics loss: Penalty for violating $\theta = \tan^{-1}\left(\frac{\mu_0 n I}{2aBh}\right)$

5.Results and Discussion

The PINN estimate agrees with the theoretical value (39.0 μT at NISER) .

The model corrects for systematic bias(it shows residuals for PINN vs. traditional method).

Ablation study: Removing physics loss increases error, confirming its necessity.

6. Conclusion

We demonstrate that PINNs enhance the precision of Earth's magnetic field measurements by fusing experimental data with physical constraints. This approach:

- 1. Reduces reliance on ideal assumptions (e.g., perfect coil geometry).**
- 2. Provides a framework for AI-augmented physics experiments.**

From our experiment we got BH is 40.28 microtesla.

Future work: Extend to 3D field mapping and real-time calibration.

References

- 1. Raissi et al. (2019). Journal of Computational Physics.**
- 2. NISER Geomagnetic Survey (2023).**

